GREEEN tallgrass prairie restoration: Seeded richness ~ Species pool size (NW Station)

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> **NOTE:** To compare different models, you only have to change the models in the section 'Load models'

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```r
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
library(lme4)
library(glmmTMB)
library(DHARMa)
library(emmeans)
```

Packages

```r
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    id_plot = "f",
    year = "f",
    seeding_time = col_factor(
      levels = c("unseeded", "fall", "spring"), ordered = FALSE
      ),
    seeded_pool = col_factor(
      levels = c("0", "6", "12", "18", "33"), ordered = TRUE
      ),
    treatment_id = "f",
    richness_type = "f"
  )
) %>%
  filter(
    site == "NW Station",
    richness_type == "seeded_richness",
    !(treatment_id %in% c("2", "4"))
  ) %>%
  mutate(y = richness_1qm + richness_25qm)
```

Load data

Statistics

Data exploration

Means and deviations

```r
Rmisc::CI(sites$y, ci = .95)
```

```
##    upper     mean    lower
## 1.682489 1.484375 1.286261
```

```r
median(sites$y)
```

```
## [1] 1
```

```r
sd(sites$y)
```

```
## [1] 1.391735
```

```r
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##  5% 95%
##   0   4
```

```r
sites %>% count(site, year)
```

```
## # A tibble: 4 x 3
##   site       year      n
##   <chr>      <fct> <int>
## 1 NW Station 2015     48
## 2 NW Station 2016     48
## 3 NW Station 2017     48
## 4 NW Station 2018     48
```

```r
sites %>% count(seeded_pool)
```

```
## # A tibble: 4 x 2
##   seeded_pool     n
##   <ord>       <int>
## 1 6              48
## 2 12             48
## 3 18             48
## 4 33             48
```

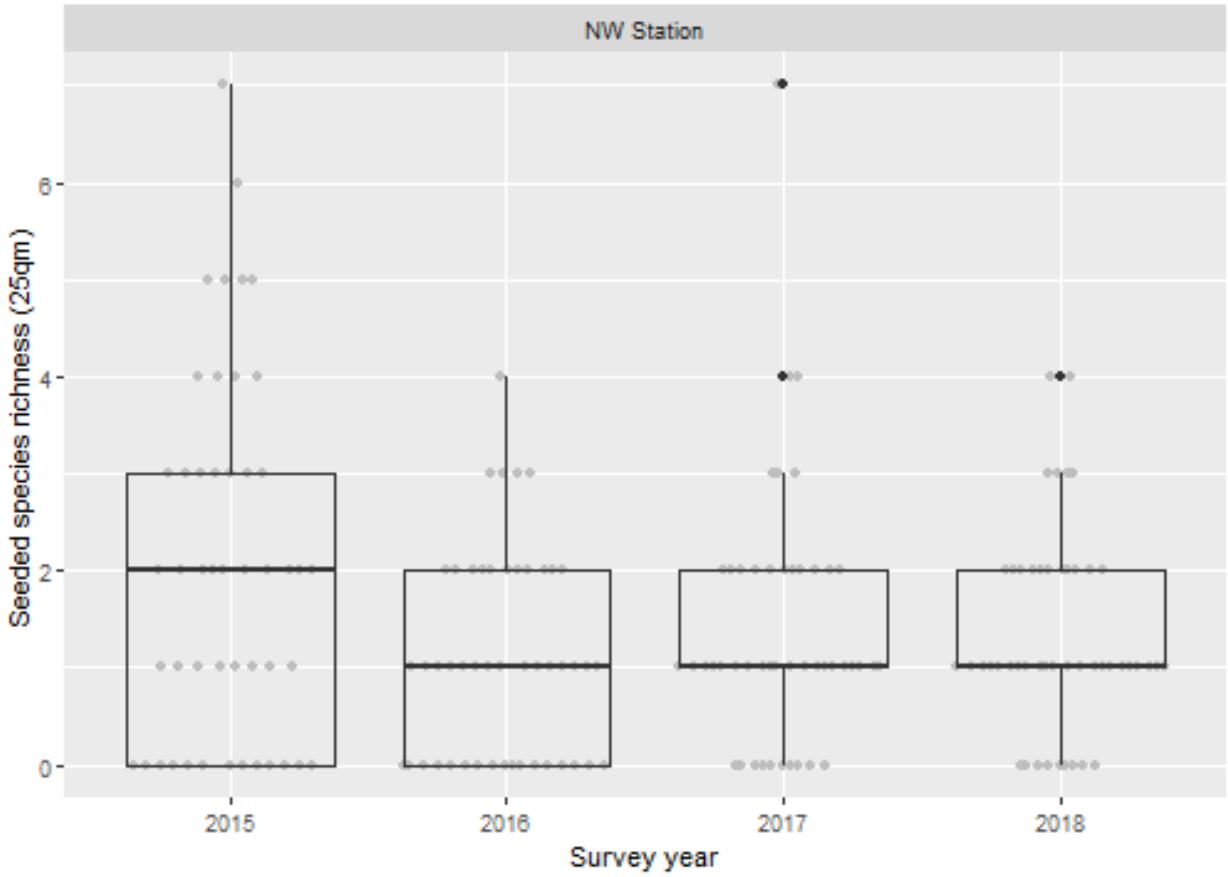
```r
sites %>% count(seeding_time)
```

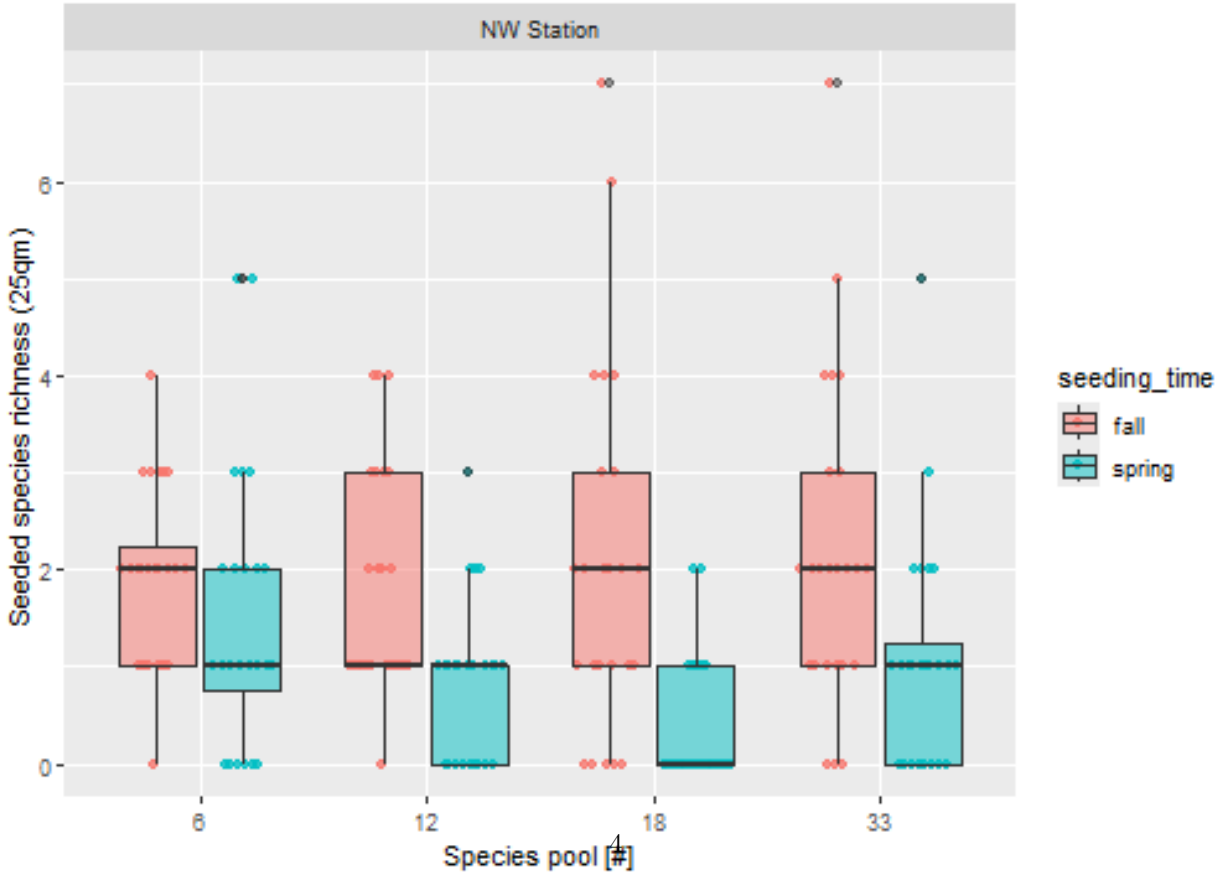
```
## # A tibble: 2 x 2
##   seeding_time     n
##   <fct>        <int>
## 1 fall            96
## 2 spring          96
```

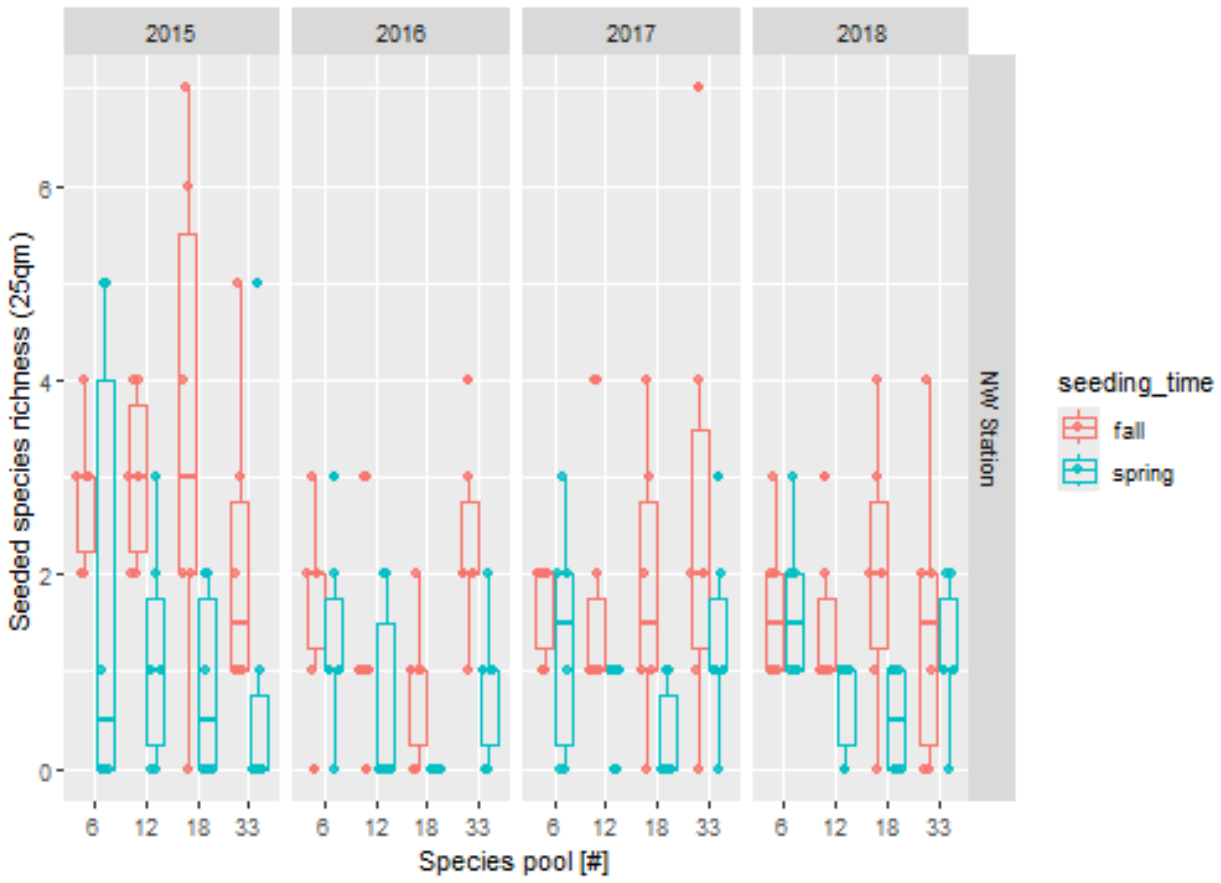
```r
sites %>% count(herbicide)
```

```
## # A tibble: 2 x 2
##   herbicide     n
##       <dbl> <int>
## 1         0    96
## 2         1    96
```

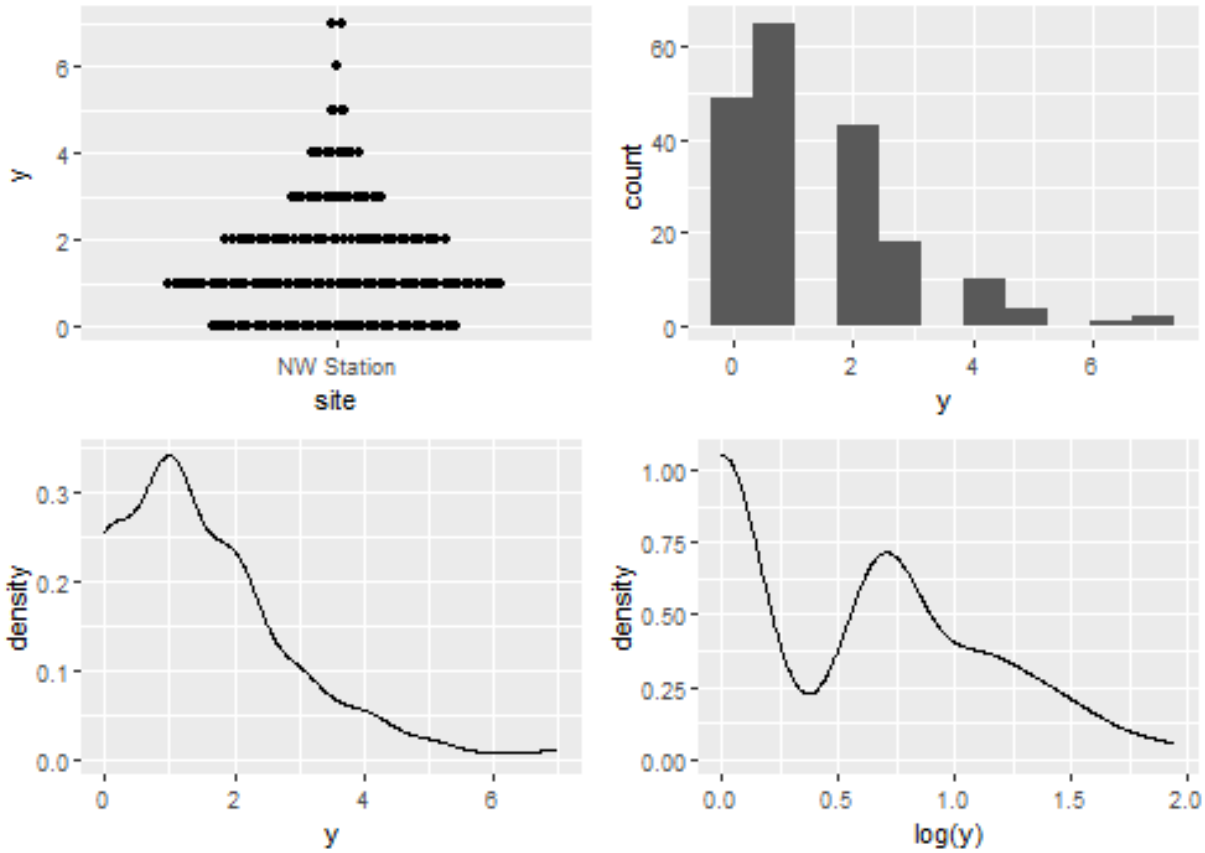
Graphs of raw data (Step 2, 6, 7)






2015
2016
2017
2018
NW Station
Seeded species richness (25qm)
0
2
4
6
6
12
18
33
Species pool [#]
seeding_time
fall
spring

Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude r > 0.7 Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cwm.")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#     ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

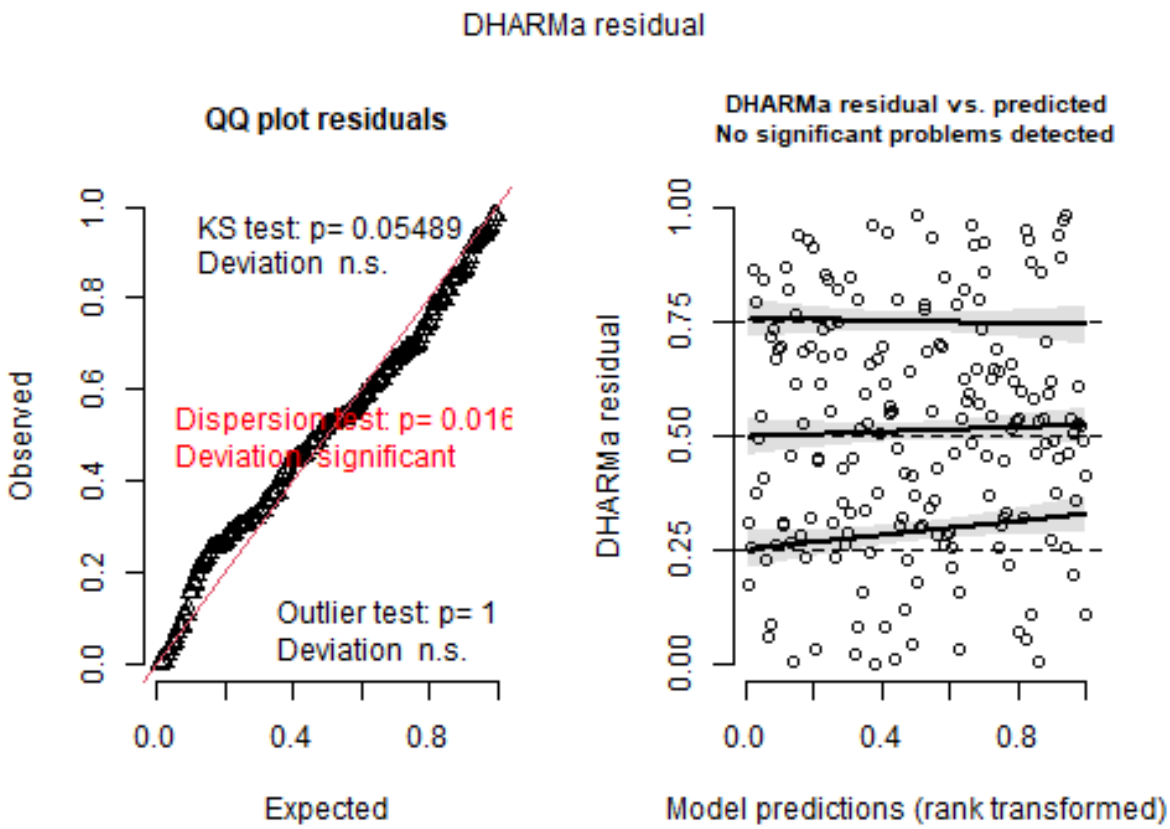
```
load(file = here("outputs", "models", "model_richness_pool_nws_1.Rdata"))
m_1 <- m1
```

```
formula(m_1)
## y ~ seeded_pool * seeding_time * year + water_cap
```

Model check

DHARMa

```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```




```
# simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```

```
# plotResiduals(simulation_output_1$scaledResiduals, factor(sites$seeded_pool))
# plotResiduals(simulation_output_2$scaledResiduals, factor(sites$seeded_pool))
# plotResiduals(simulation_output_1$scaledResiduals, sites$herbicide)
# plotResiduals(simulation_output_2$scaledResiduals, sites$herbicide)
# plotResiduals(simulation_output_1$scaledResiduals, sites$seeding_time)
# plotResiduals(simulation_output_2$scaledResiduals, sites$seeding_time)
# plotResiduals(simulation_output_1$scaledResiduals, sites$year)
# plotResiduals(simulation_output_2$scaledResiduals, sites$year)
# plotResiduals(simulation_output_1$scaledResiduals, sites$water_cap)
# plotResiduals(simulation_output_2$scaledResiduals, sites$water_cap)
```

Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
## there are higher-order terms (interactions) in this model
## consider setting type = 'predictor'; see ?vif
```

```
##                                     GVIF Df GVIF^(1/(2*Df))
## seeded_pool                   7.231685e+01  3        2.041142
## seeding_time                  3.292726e+00  1        1.814587
## year                          3.906299e+00  3        1.254953
## water_cap                     1.247484e+00  1        1.116908
## seeded_pool:seeding_time      8.522335e+01  3        2.097779
## seeded_pool:year              9.004296e+02  9        1.459271
## seeding_time:year             1.199527e+07  3       15.129864
## seeded_pool:seeding_time:year 1.619733e+09  9        3.248148
```

```
# car::vif(m_2)
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
## Warning: the null model is only correct if all the variables it uses are identical
## to those used in fitting the original model.
##                 R2m       R2c
## delta     0.9366897 0.9366897
## lognormal 0.9508671 0.9508671
## trigamma  0.9131556 0.9131556
# MuMIn::r.squaredGLMM(m_2)
```

AICc

Use AICc and not AIC since ratio n/K < 40 Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
# MuMIn::AICc(m_1, m_2) %>%
#   arrange(AICc)
```

Predicted values

Summary table

```
car::Anova(m_1, type = 3)
```

```
## Analysis of Deviance Table (Type III tests)
##
## Response: y
##                               LR Chisq Df Pr(>Chisq)
## seeded_pool                     1.9486  3   0.583151
## seeding_time                   21.1966  1  4.145e-06 ***
## year                           13.3646  3   0.003911 **
## water_cap                      27.8979  1  1.279e-07 ***
## seeded_pool:seeding_time        3.8014  3   0.283724
## seeded_pool:year                8.8768  9   0.448723
## seeding_time:year               2.4469  3   0.484972
## seeded_pool:seeding_time:year   3.3694  9   0.947837
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(m_1)
```

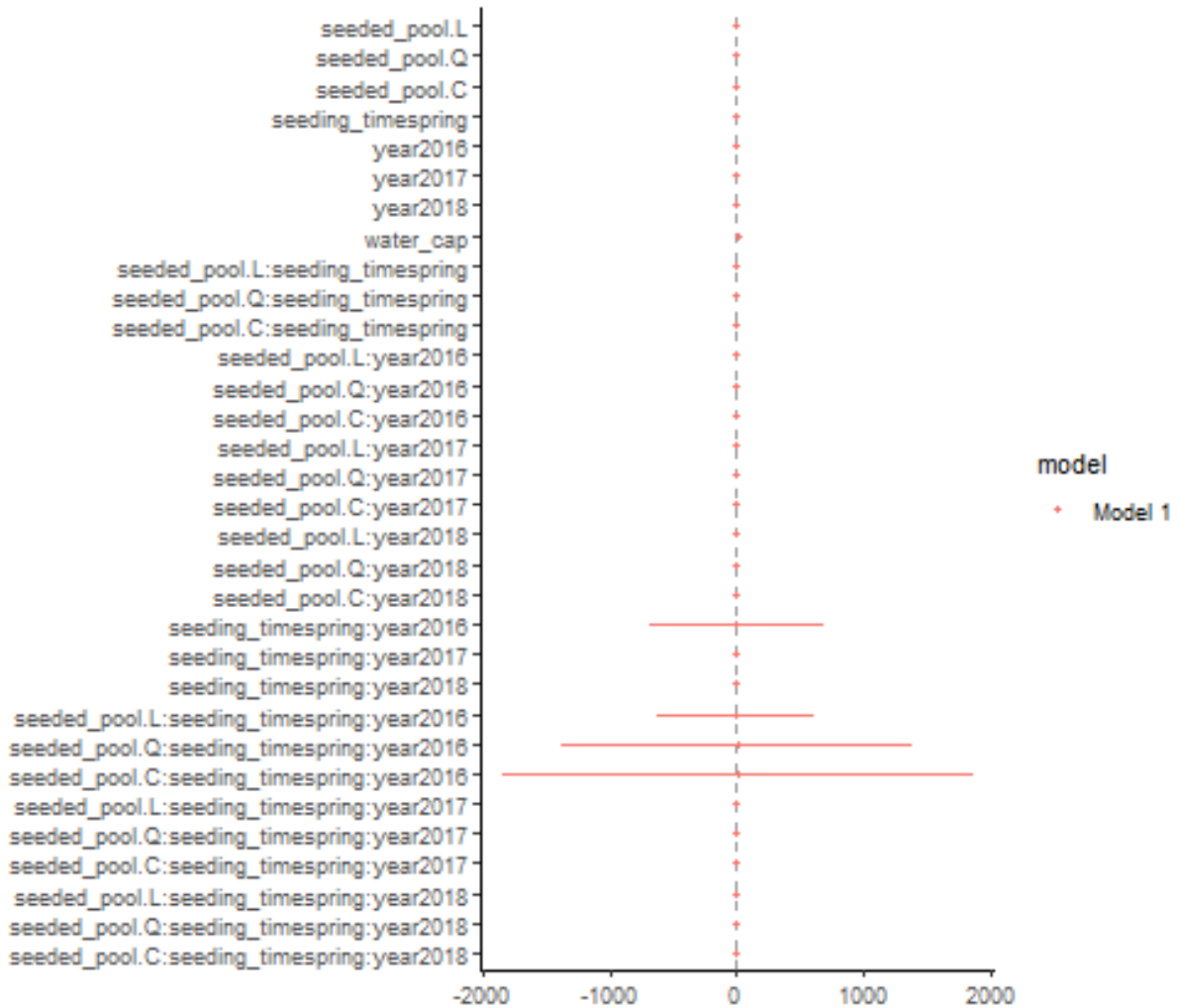
```
##
```

```
## Call:
## glm(formula = y ~ seeded_pool * seeding_time * year + water_cap,
##     family = poisson(link = "log"), data = sites)
##
## Coefficients:
##                                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)                              -2.29728    0.63000  -3.646 0.000266
## seeded_pool.L                            -0.01617    0.25816  -0.063 0.950072
## seeded_pool.Q                            -0.33568    0.24448  -1.373 0.169736
## seeded_pool.C                            -0.03821    0.23327  -0.164 0.869885
## seeding_timespring                       -1.00601    0.23094  -4.356 1.32e-05
## year2016                                 -0.70902    0.21697  -3.268 0.001084
## year2017                                 -0.38935    0.19241  -2.024 0.043014
## year2018                                 -0.53778    0.20070  -2.680 0.007372
## water_cap                                 9.81550    1.79168   5.478 4.29e-08
## seeded_pool.L:seeding_timespring         -0.19380    0.44671  -0.434 0.664405
## seeded_pool.Q:seeding_timespring          0.80991    0.45905   1.764 0.077681
## seeded_pool.C:seeding_timespring         -0.10337    0.47966  -0.216 0.829375
## seeded_pool.L:year2016                    0.29596    0.40066   0.739 0.460099
## seeded_pool.Q:year2016                    0.96151    0.43394   2.216 0.026706
## seeded_pool.C:year2016                    0.46434    0.46484   0.999 0.317825
## seeded_pool.L:year2017                    0.48209    0.38590   1.249 0.211566
## seeded_pool.Q:year2017                    0.45571    0.38481   1.184 0.236319
## seeded_pool.C:year2017                    0.20455    0.38373   0.533 0.593990
## seeded_pool.L:year2018                    0.13914    0.41348   0.337 0.736492
## seeded_pool.Q:year2018                    0.17720    0.40140   0.441 0.658872
## seeded_pool.C:year2018                   -0.05315    0.38894  -0.137 0.891305
## seeding_timespring:year2016              -3.88828  351.36190  -0.011 0.991171
## seeding_timespring:year2017               0.03508    0.36748   0.095 0.923939
## seeding_timespring:year2018               0.28488    0.35978   0.792 0.428473
## seeded_pool.L:seeding_timespring:year2016  -4.05414  314.26823  -0.013 0.989707
## seeded_pool.Q:seeding_timespring:year2016   7.50917  702.72380   0.011 0.991474
## seeded_pool.C:seeding_timespring:year2016  10.61562  942.80271   0.011 0.991016
## seeded_pool.L:seeding_timespring:year2017  -0.29471    0.66144  -0.446 0.655920
## seeded_pool.Q:seeding_timespring:year2017   0.08852    0.73495   0.120 0.904135
## seeded_pool.C:seeding_timespring:year2017   0.29008    0.80175   0.362 0.717497
## seeded_pool.L:seeding_timespring:year2018   0.03912    0.66519   0.059 0.953108
## seeded_pool.Q:seeding_timespring:year2018   0.38744    0.71957   0.538 0.590280
## seeded_pool.C:seeding_timespring:year2018   0.07620    0.77012   0.099 0.921180
##
## (Intercept)                              ***
## seeded_pool.L
## seeded_pool.Q
## seeded_pool.C
## seeding_timespring                       ***
## year2016                                 **
## year2017                                 *
## year2018                                 **
## water_cap                                ***
## seeded_pool.L:seeding_timespring
## seeded_pool.Q:seeding_timespring         .
## seeded_pool.C:seeding_timespring
## seeded_pool.L:year2016
## seeded_pool.Q:year2016                   *
```

```
## seeded_pool.C:year2016
## seeded_pool.L:year2017
## seeded_pool.Q:year2017
## seeded_pool.C:year2017
## seeded_pool.L:year2018
## seeded_pool.Q:year2018
## seeded_pool.C:year2018
## seeding_timespring:year2016
## seeding_timespring:year2017
## seeding_timespring:year2018
## seeded_pool.L:seeding_timespring:year2016
## seeded_pool.Q:seeding_timespring:year2016
## seeded_pool.C:seeding_timespring:year2016
## seeded_pool.L:seeding_timespring:year2017
## seeded_pool.Q:seeding_timespring:year2017
## seeded_pool.C:seeding_timespring:year2017
## seeded_pool.L:seeding_timespring:year2018
## seeded_pool.Q:seeding_timespring:year2018
## seeded_pool.C:seeding_timespring:year2018
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 257.64  on 187  degrees of freedom
## Residual deviance: 146.63  on 155  degrees of freedom
##   (4 Beobachtungen als fehlend gelöscht)
## AIC: 562.73
##
## Number of Fisher Scoring iterations: 15
```

Forest plot

```r
dotwhisker::dwplot(
  list(m_1),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  theme_classic()
```



Effect sizes

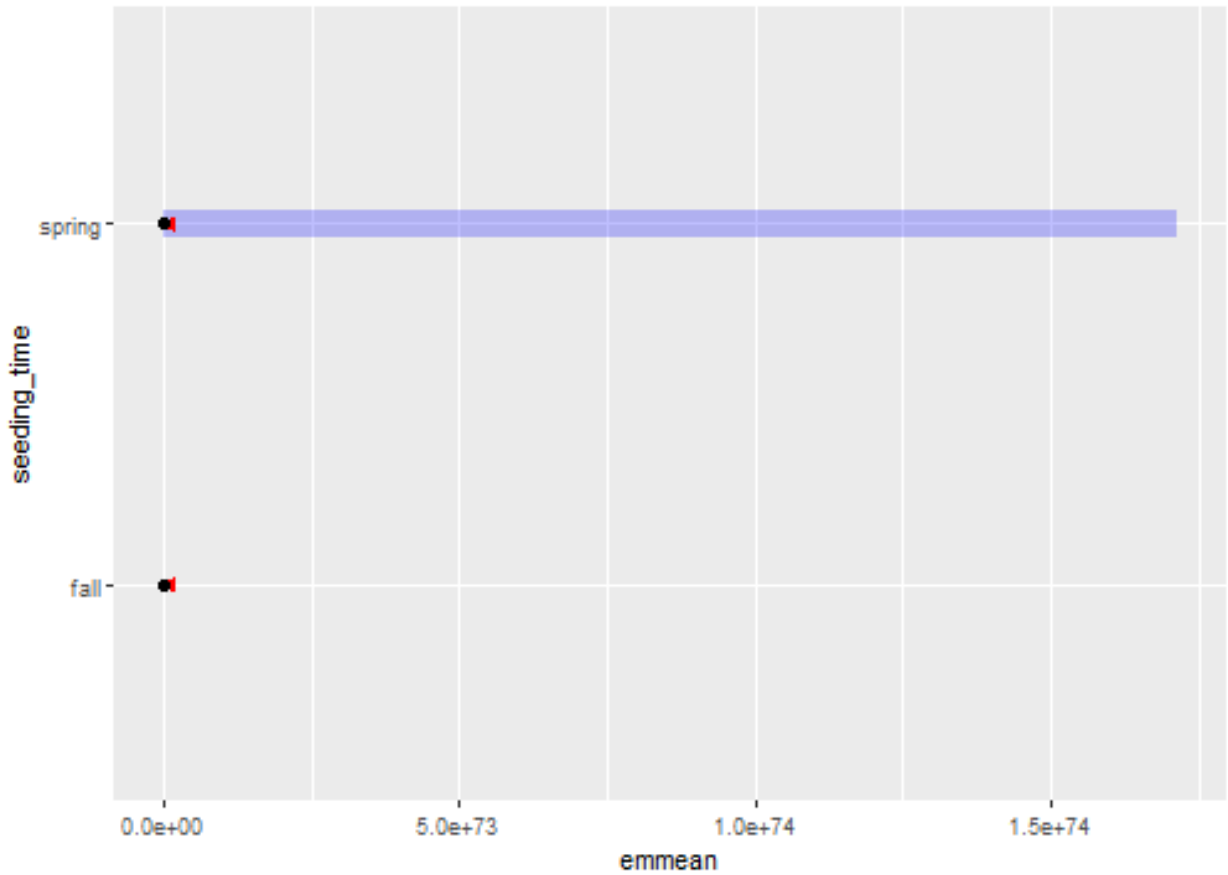
Effect sizes of chosen model just to get exact values of means etc. if necessary.

```
(emm <- emmeans(
  m_1,
  revpairwise ~ seeding_time,
  type = "response"
  ))
```

```
## $emmeans
##  seeding_time  rate     SE  df asymp.LCL asymp.UCL
##  fall         1.940  0.149 Inf      1.67  2.00e+00
##  spring       0.291 25.500 Inf      0.00  1.71e+74
##
## Results are averaged over the levels of: seeded_pool, year
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
##
## $contrasts
```

```
##  contrast      ratio   SE  df null z.ratio p.value
##  spring / fall  0.15 13.2 Inf    1  -0.022  0.9828
##
## Results are averaged over the levels of: seeded_pool, year
## Tests are performed on the log scale
```

```
plot(emm, comparison = TRUE)
```



Session info

```
## R version 4.5.0 (2025-04-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
```

```
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_1.11.1   DHARMa_0.4.7     glmmTMB_1.1.12   lme4_1.1-37
##  [5] Matrix_1.7-3     patchwork_1.3.1  ggbeeswarm_0.7.2 lubridate_1.9.4
##  [9] forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4      purrr_1.1.0
## [13] readr_2.1.5      tidyr_1.3.1      tibble_3.3.0     ggplot2_3.5.2
## [17] tidyverse_2.0.0  here_1.0.1
##
## loaded via a namespace (and not attached):
##  [1] Rdpack_2.6.4           gridExtra_2.3          sandwich_3.1-1
##  [4] rlang_1.1.6            magrittr_2.0.3         compiler_4.5.0
##  [7] mgcv_1.9-3             vctrs_0.6.5            pkgconfig_2.0.3
## [10] crayon_1.5.3           fastmap_1.2.0          backports_1.5.0
## [13] labeling_0.4.3         utf8_1.2.6             ggstance_0.3.7
## [16] promises_1.3.3         rmarkdown_2.29         tzdb_0.5.0
## [19] nloptr_2.2.1           bit_4.6.0              xfun_0.52
## [22] later_1.4.2            parallel_4.5.0         R6_2.6.1
## [25] gap.datasets_0.0.6     stringi_1.8.7          qgam_2.0.0
## [28] RColorBrewer_1.1-3     car_3.1-3              boot_1.3-31
## [31] numDeriv_2016.8-1.1    estimability_1.5.1     Rcpp_1.1.0
## [34] iterators_1.0.14       knitr_1.50             zoo_1.8-14
## [37] parameters_0.27.0      httpuv_1.6.16          splines_4.5.0
## [40] timechange_0.3.0       tidyselect_1.2.1       rstudioapi_0.17.1
## [43] abind_1.4-8            yaml_2.3.10            MuMIn_1.48.11
## [46] doParallel_1.0.17      TMB_1.9.17             codetools_0.2-20
## [49] lattice_0.22-7         plyr_1.8.9             shiny_1.11.1
## [52] withr_3.0.2            bayestestR_0.16.1      coda_0.19-4.1
## [55] evaluate_1.0.4         marginaleffects_0.28.0 pillar_1.11.0
## [58] gap_1.6                carData_3.0-5          foreach_1.5.2
## [61] stats4_4.5.0           reformulas_0.4.1       insight_1.3.1
## [64] generics_0.1.4         vroom_1.6.5            rprojroot_2.0.4
## [67] hms_1.1.3              scales_1.4.0           minqa_1.2.8
## [70] xtable_1.8-4           glue_1.8.0             tools_4.5.0
## [73] data.table_1.17.8      mvtnorm_1.3-3          grid_4.5.0
## [76] rbibutils_2.3          datawizard_1.1.0       nlme_3.1-168
## [79] Rmisc_1.5.1            performance_0.15.0     beeswarm_0.4.0
## [82] vipor_0.4.7            Formula_1.2-5          cli_3.6.5
## [85] gtable_0.3.6           digest_0.6.37          farver_2.1.2
## [88] htmltools_0.5.8.1      lifecycle_1.0.4        mime_0.13
## [91] bit64_4.6.0-1          dotwhisker_0.8.4       MASS_7.3-65
```